DK A DATA STORY FOR NHS POLICY MAKERS

The appointments *nobody* attends

What 200,000 GP appointment records reveal about who misses appointments in England, and why two common assumptions are not supported by the data.

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NHS England · DLUHC/ONS open data

July 2022 – December 2024

Missed GP appointments cost the NHS more than £1 billion a year

Every “did not attend” (DNA) is a slot another patient needed. The question for policy makers is where the problem is concentrated, and what drives it.

£1bn+

estimated annual cost of missed GP appointments in England

200k+

appointment records analysed, across 96 Sub-ICBs and 7 NHS regions

30

months of NHS England's own published data, July 2022 – December 2024

A stable national average hides a wide regional spread

A stable national temperature can coexist with a heatwave in one city and frost in another. DNA rates behave the same way.

NATIONAL DNA RATE, 30 MONTHS

4.8%

Broadly flat between 4.4% and 5.9%, with no sign of a national crisis in the trend line.

0–33% the actual spread of DNA rates across England's 96 Sub-ICB areas

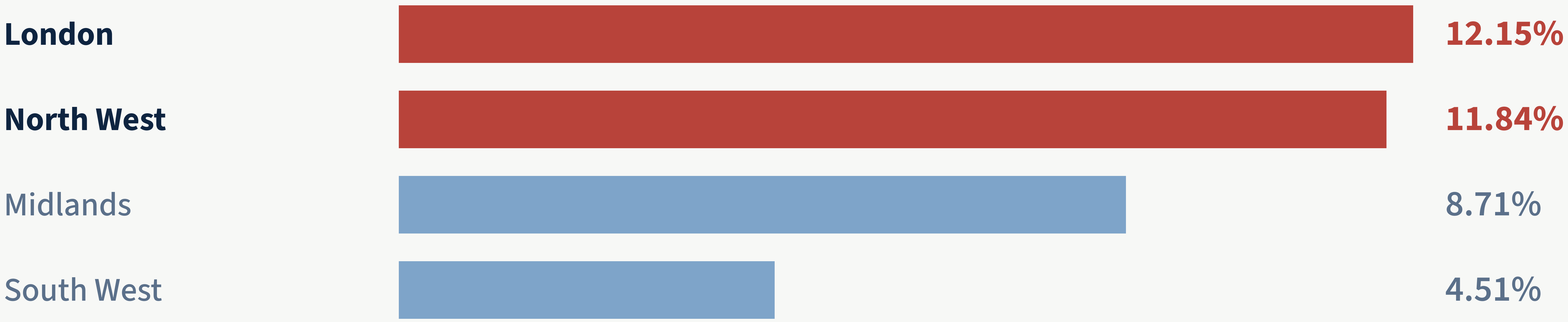
~1 in 2 Sub-ICBs run above 8%, well above what the national figure suggests

2.7× gap between the highest region (London) and the lowest (South West)

FINDING 1

London and the North West sit in a tier of their own

High-rate areas cluster inside two regions, a pattern that points to structural and operational factors, not individual patient behaviour.



Five of the ten worst Sub-ICBs nationally sit in the North West alone. The East of England, by contrast, supplies most of the best performers, with several below 3.5%.

FINDING 2

Face-to-face appointments are missed five times more often than video

The common assumption that remote appointments are harder to attend runs the wrong way. The pattern holds in all seven NHS regions.

10.65%	5.77%	2.68%	1.94%
Face-to-face	**Telephone**	**Home visit**	**Video / online**
11.5M appointments	1.8× lower	4× lower	5.5× lower

The likeliest mechanism is **cancellation friction** rather than travel cost: a face-to-face slot carries a stronger social commitment, so when patients abandon one they tend to do so silently, as a DNA rather than a cancellation.

The shift to remote care has reduced missed appointments

Video and online consultation is the fastest-growing mode in general practice, and also the one patients miss least.

10×

growth in video/online share in 2.5 years, from 0.58% to 5.54% of all appointments

<6%

of appointments are video/online today; face-to-face still carries ~67% of all volume

0.3–0.5pp

estimated fall in the national DNA rate if video/online share doubled, equal to tens of thousands of avoided DNAs a year

Clinical appropriateness remains the allocation criterion. The opportunity is in appointments already suitable for remote delivery, such as routine follow-ups and reviews.

FINDING 3

Deprived areas miss more appointments, but the signal is too noisy to target on

The most deprived Sub-ICBs average the highest DNA rate. But the relationship is not linear, and the spread within every band is enormous.

MOST DEPRIVED BAND (20 AREAS)

11.24%

average DNA rate, with Lancashire (13.65%), Greater Manchester (14.88%) and Cheshire (16.93%) all in this band.

The pattern is not linear

Mid-deprivation bands run 8.1–8.8%, out of order. Deprivation alone doesn't predict DNA rates.

Within-band spread is huge

One band spans 0% to 32.8%, so the average describes no actual Sub-ICB.

A hypothesis, not a targeting rule

Mode mix and urban density are plausible confounders. Controlled analysis first; resource allocation after.

Three implications for policy

1

Grow the lowest-friction modes

Expanding video and telephone access for clinically suitable appointments reduces DNAs without asking patients to change behaviour. Mode-share ambitions could sit at Sub-ICB level.

2

Focus on the hotspot regions

The concentration in London and the North West points to regional review of mode mix, reminder systems and booking lead times, rather than uniform national programmes.

3

Monitor deprivation; validate before acting

DNA-by-deprivation belongs in routine monitoring, published alongside mode mix, so that a friction problem is not misread as patient behaviour in deprived areas.

The framing may matter as much as the finding: this looks less like an "access inequality" story and more like a "friction and commitment" story, with different interventions to match.

What this analysis can and cannot say

These findings are strong and consistent, but they are observational. They support controlled follow-up, not causal claims.

Booking lead time is unmeasured

Same-day telephone triage and advance-booked face-to-face slots have structurally different DNA dynamics.

Patients select into modes

Clinically complex patients are directed to face-to-face, complicating direct mode comparison.

Coverage is PCN-hub systems

The granular data may not represent every GP appointment in each Sub-ICB area.

Small volumes distort extremes

The 0% and 32.8% outliers rest on very few appointments; deprivation data dates from 2019.

About this analysis

A capstone portfolio project for the Google Data Analytics Professional Certificate. The full case study documents the SQL cleaning decisions, the corrected LSOA-to-Sub-ICB geographic join, and every limitation behind these findings.

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TOOLS & METHOD

SQL (SQLite) for cleaning & analysis

Microsoft Excel for visualisation

6 documented cleaning decisions

3 datasets joined across geographies

DATA SOURCES

Appointments in General Practice · NHS England

English Indices of Deprivation 2019 · DLUHC / ONS

Open Government Licence v3.0 · no patient-identifiable data